

MK
KS
ML
AB
HT
CF
MB
GS
AM/YR
DH
WS

NAME _____

CLASS 12MT____ **or 12MTX**____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2018

YEAR 12

AP4

MATHEMATICS

Time allowed – 3 HOURS + 5 Minutes Reading Time

DIRECTIONS TO CANDIDATES:

SECTION I: Use the multiple-choice answer sheet provided.

SECTION II:

- Each question is to be commenced in a new booklet clearly marked Question 11, Question 12, etc on the top of the page.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- If you do not attempt a question, you must submit a blank booklet clearly indicating the question number, your name and class
- All questions should be placed in order - multiple-choice answer sheet followed by Question 11 to 16.
- NESAs approved calculators may be used.
- The NESAs Reference Sheet is provided.

Section I

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

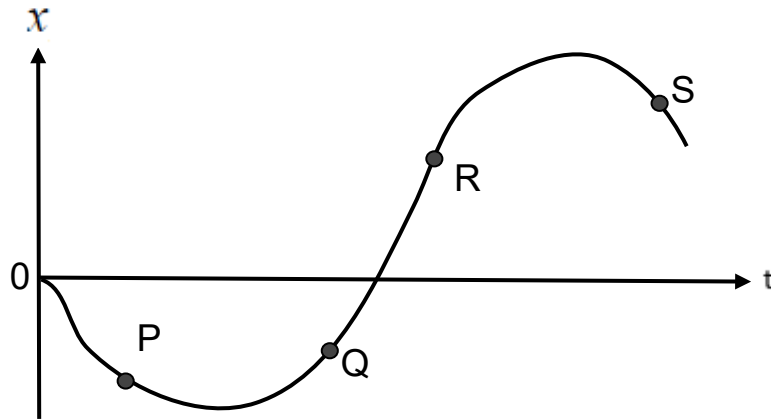
Use the multiple-choice answer sheet for Questions 1 - 10

1. Which of the following is a primitive of $\frac{2}{x} - \sin x$?
 - (A) $2 \ln|x| + \cos x + C$
 - (B) $2 \ln|x| - \cos x + C$
 - (C) $-\frac{2}{x^2} + \cos x + C$
 - (D) $-\frac{2}{x^2} - \cos x + C$
2. What is the value of $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$?
 - (A) 0
 - (B) 4
 - (C) 7
 - (D) 10
3. The domain of the function $f(x) = \frac{x}{\sqrt{x+3}}$ is:
 - (A) $x \neq -3$
 - (B) $x > 3$
 - (C) $x > -3$
 - (D) $x \geq -3$
4. Which expression is a term of the geometric series $4x - 8x^2 + 16x^3 - \dots$?
 - (A) $-2048 x^9$
 - (B) $-2048 x^{10}$
 - (C) $2048 x^9$
 - (D) $2048 x^{10}$
5. What is the equation of the tangent to the curve $y = \sin x$ at the point $(\pi, 0)$?
 - (A) $y = 0$
 - (B) $x - y + \pi = 0$
 - (C) $x - y - \pi = 0$
 - (D) $x + y - \pi = 0$

6. What is the value of k if the equation $3x^2 + kx - 4 = 0$ has -2 as one of its roots?

- (A) $k = 4$
- (B) $k = 8$
- (C) $k = -4$
- (D) $k = -8$

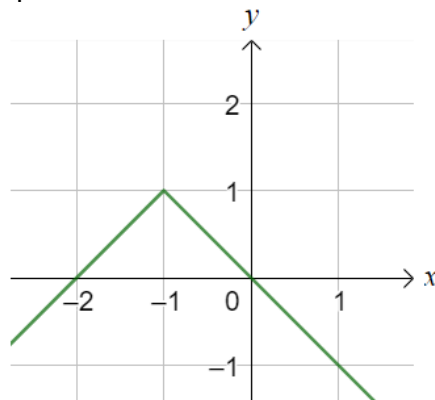
7. The displacement-time graph of a particle moving in a straight line is as shown.



At which point is the velocity negative and the particle is slowing down?

- (A) P
- (B) Q
- (C) R
- (D) S

8. What is the equation of the graph below?

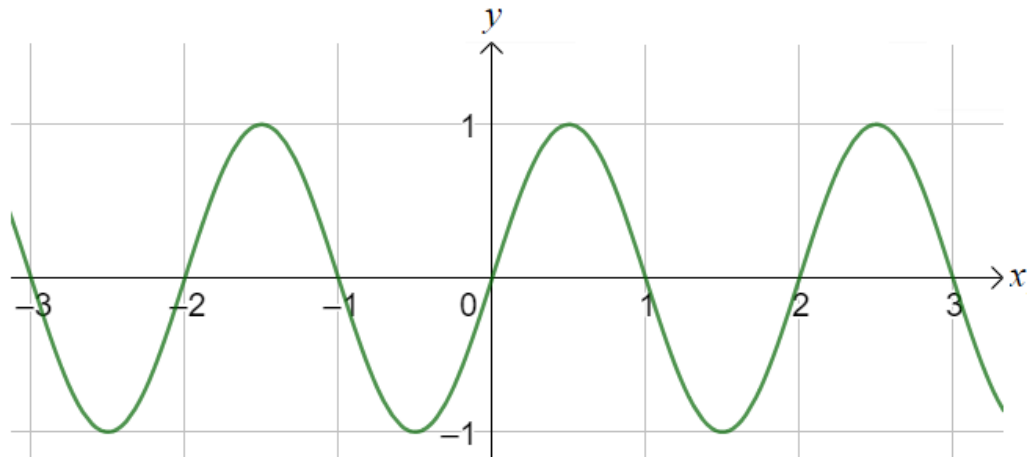


- (A) $y = |x - 1| - 1$
- (B) $y = |x + 1| - 1$
- (C) $y = 1 - |x - 1|$
- (D) $y = 1 - |x + 1|$

9. Which equation represents an odd function?

- (A) $y = \sin x + \cos x$
- (B) $y = \sin x - \cos x$
- (C) $y = \tan x + \sec x$
- (D) $y = \tan x + \operatorname{cosec} x$

10. The graph of the function $y = \sin(\pi x)$ is given below.



How many solutions does $2\sin(\pi x) + x = 0$ have?

- (A) 5
- (B) 3
- (C) 1
- (D) None of the above

End of Section I

Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Start a new writing booklet	Marks
(a) Factorise $4x^2 + 11x - 3$		1
(b) If $\frac{8}{5 - \sqrt{2}}$ equals $a + b\sqrt{2}$. Find the values of a and b .		2
(c) Solve $ 2x - 1 = 5$		2
(d) Find the exact value of $\tan \frac{\pi}{6} + \sec \frac{\pi}{4}$		2
(e) For the arithmetic sequence 31, 27, 23, 19,		
(i) What is the 30th term?		1
(ii) Find the sum of the first 50 terms.		1
(f) The equation $2x^2 + 5x - 4 = 0$ has roots α and β . Find the values of		
(i) $\alpha + \beta$		1
(ii) $\alpha\beta$		1
(iii) $(\alpha - 1)(\beta - 1)$		1
(g)		
(i) Sketch the graph of $y = x^2 - 9$, showing all intercepts.		2
(ii) Find the area between the curve $y = x^2 - 9$ and the x -axis.		1

End of Question 11

Question 12 (15 marks)

Start a new writing booklet

Marks

(a) For the parabola $x^2 - 6x - 8y - 31 = 0$,

(i) Show that $x^2 - 6x - 8y - 31 = 0$ can be written as

$(x - 3)^2 = 8(y + 5)$ by completing the square.

(ii) Find the focal length.

(iii) Find the coordinates of the focus.

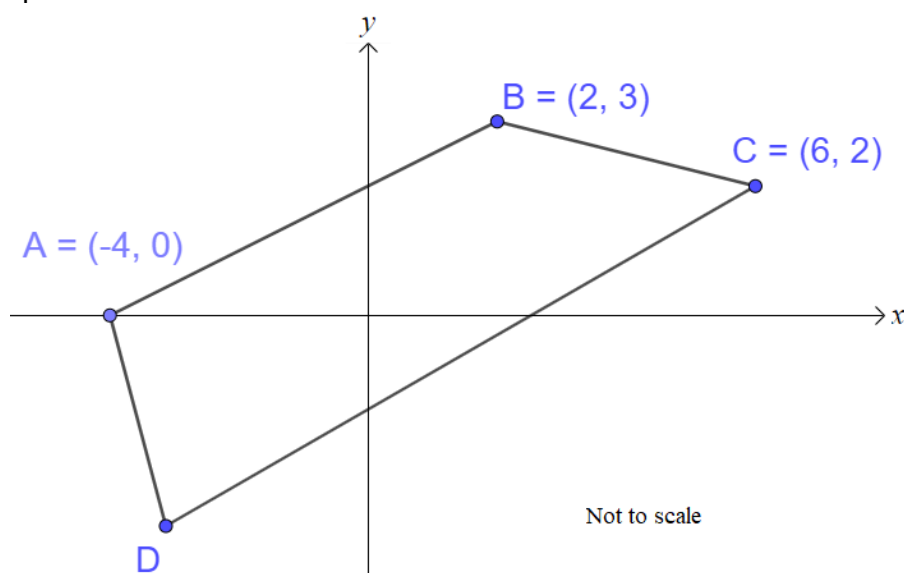
(iv) Find the equation of the directrix.

(v) Find the equation of the normal to the parabola at the point $(11, 3)$.

(b) The graph of the function $y = f(x)$ passes through the point $(1, 6)$.

If $f'(x) = 4x - 5$, find $f(x)$.

(c) In the diagram, the points $A(-4, 0)$, $B(2, 3)$, $C(6, 2)$ and D are vertices of a trapezium.



(i) Show that the equation of the line CD is $x - 2y - 2 = 0$.

(ii) Given the equation of the line AD is $x + y + 4 = 0$, show that the coordinates of point D are $(-2, -2)$.

(iii) Find the length AB. Give your answer in exact form.

(iv) Find the perpendicular distance from point A to the line CD.

(v) Find the area of the trapezium ABCD if length $CD = \sqrt{80}$.

End of Question 12

Question 13 (15 marks)

Start a new writing booklet

Marks

(a) Solve $4\sin^2 2x = 3$ for $-\pi \leq x \leq \pi$

3

(b) The function $y = f(x)$ is defined by $f(x) = x^3 + 6x^2 + 9x + 1$.

(i) Find any turning points and determine their nature.

3

(ii) Find any point(s) of inflexion.

1

(iii) Sketch the curve, showing all turning points and point(s) of inflexion.

1

(c) The table below gives some values for $f(x) = x \log_e x$

x	1	2	3	4	5
$f(x)$	0		3.33		8.05

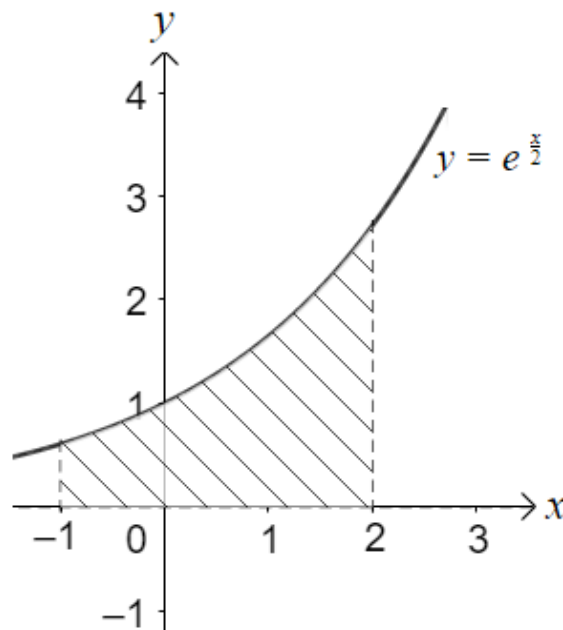
(i) Copy and complete the table of values in your writing booklet.

1

(ii) Use Simpson's Rule with these 5 values to find an approximation of

2

$$\int_1^5 x \log_e x \, dx \quad \text{correct to 2 decimal places.}$$

(d) The region bounded by the graph of $y = x^2$, the x -axis and $x = 2$ is rotated about the x -axis. Find the volume of the solid of revolution formed.**2**(e) The graph of $y = e^{\frac{x}{2}}$ is shown below.**2**

The shaded region is bounded by $y = e^{\frac{x}{2}}$, the x -axis, the lines $x = -1$ and $x = 2$. Find the area of the shaded region, correct to 2 decimal places.

End of Question 13

Question 14 (15 marks)

Start a new writing booklet

Marks

- (a) Mary borrows \$50 000 at 6% compound interest per annum from her bank. She makes \$4000 repayments each year. The amount owed at the end of the n th year is $\$A_n$.

(i) Show that $A_2 = 50\,000(1.06)^2 - (4000(1.06) + 4000)$

1

(ii) Show that $A_n = 50\,000(1.06)^n - \frac{4000(1.06^n - 1)}{0.06}$

2

- (iii) Find the amount of money Mary owes at the end of 12 years. Give your answer correct to the nearest dollar.

1

- (b) If $\log_m a = 1.2$ and $\log_m b = 0.5$, find the value of

(i) $\log_m(ab)$

1

(ii) $\log_m\left(\frac{b^3}{a}\right)$

2

- (c)

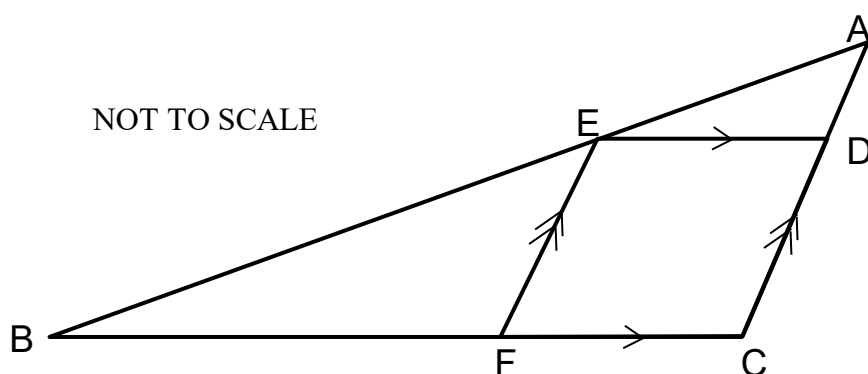
(i) Show that $\frac{1}{2x-3} - \frac{1}{2x+3} = \frac{6}{4x^2-9}$

1

(ii) Hence find $\int \frac{dx}{4x^2-9}$ and fully simplify your answer.

3

- (d)



In the diagram above, CDEF is a rhombus inscribed inside the triangle ABC.

- (i) Prove that $\triangle ADE$ is similar to $\triangle EFB$.

2

- (ii) Let the side of the rhombus be x units, length $BC = 12$ units and $AD = 2$ units. Find the value of x .

2**End of Question 14**

Question 15 (15 marks)

Start a new writing booklet

Marks

- (a) A tank initially containing 500 litres of water, is leaking.

The rate at which water is leaking from the tank is given by $\frac{dV}{dt} = -\frac{1600}{(t+4)^2}$

where V is the volume of water remaining in the tank and t is the time in hours after the water started to leak.

- (i) Find the volume of the water in the tank after 20 hours.

3

- (ii) How much water will be left in the tank eventually?

1

- (b) A number of frogs were introduced to an island. The population of frogs after t months is given by $P(t) = Ae^{kt}$, where A and k are constants.

- (i) Show that $P(t)$ satisfies $\frac{dP}{dt} = kP$.

1

- (ii) Five months after the frogs were introduced, their population is now 80% **more** than the initial population.

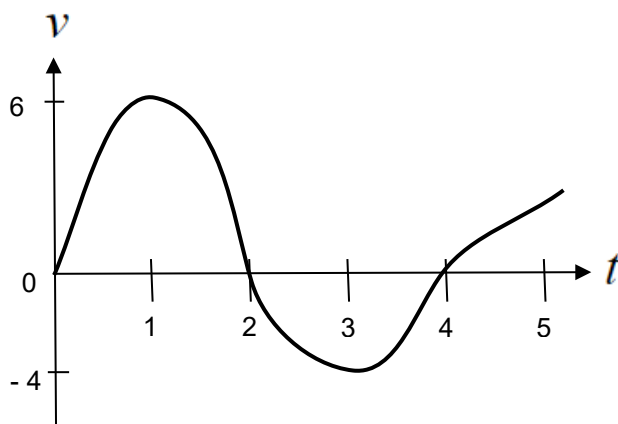
2

Show that $k = 0.1176$ correct to 4 significant figures.

- (iii) After how many months will the number of frogs first exceed one hundred times its initial population?

2

- (c)



The graph above shows the velocity v , in m/s of a particle after t seconds moving in a straight line.

- (i) What is the acceleration when $t = 1$ second?

1

- (ii) When does the particle first change direction?

1

- (iii) What does the area under the curve from $t = 0$ to $t = 2$ represent?

1

- (d) A particle that is initially at rest at the origin starts moving along the x -axis.

The velocity of the particle at time t is given by $\dot{x} = 12(t - t^3)$ ms⁻¹.

- (i) Once moving, when does the particle come to rest again?

1

- (ii) Find the distance travelled in the first 2 seconds.

2

End of Question 15

Question 16 (15 marks)

Start a new writing booklet

Marks(a) Differentiate with respect to x

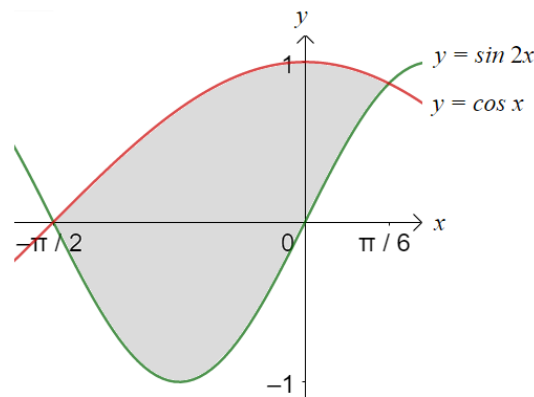
(i) $\tan \frac{x}{3}$

1

(ii) $\sin^4 x$

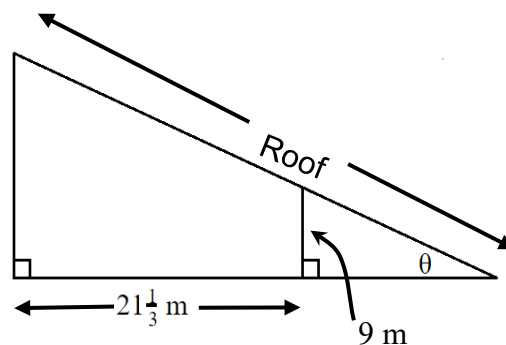
1(b) Find $\int \sec^2\left(\frac{x-3}{2}\right) dx$ **1**(c) (i) Given $y = \sin x^2$, find y' **1**(ii) Hence, or otherwise, determine $\int x \cos x^2 dx$ **2**

(d)

3

The diagram shows the graphs of $y = \cos x$ and $y = \sin 2x$. The two graphs intersect at $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{6}$. Calculate the area of the shaded regions.

(e)



A structure is made of two vertical walls and a roof, making an angle θ with the ground. The horizontal distance between the two walls is $21\frac{1}{3}$ metres and the height of the shorter wall is 9 metres.

(i) Show that the length of the roof is $L = \frac{9}{\sin \theta} + \frac{64}{3 \cos \theta}$ **3**

(ii) Find the length of the shortest roof.

3**End of Paper**

HSC Mathematics

AP4 2018

NAME _____

CLASS 12 MA _____ 12 MTX _____

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Question	Marks
1-10	/ 10
11	/ 15
12	/ 15
13	/ 15
14	/ 15
15	/ 15
16	/ 15
Total	/ 100

1) A 2) C 3) C 4) B 5) D 6) A 7) A 8) D 9) D 10) A

11) a) $4x^2 + 11x - 3$

$$= (4x - 1)(x + 3)$$

b) $\frac{8(5 + \sqrt{2})}{(5 - \sqrt{2})(5 + \sqrt{2})}$

$$= \frac{40 + 8\sqrt{2}}{23}$$

$$= \frac{40}{23} + \frac{8\sqrt{2}}{23}$$

$$a = \frac{40}{23}, b = \frac{8}{23}$$

c) $2x - 1 = \pm 5$

$$2x = -4 \quad \text{or} \quad 2x = 6$$

$$x = -2 \quad x = 3$$

d) $\frac{1}{\sqrt{3}} + \sqrt{2}$

$$= \frac{\sqrt{3}}{3} + \sqrt{2}$$

$$= \frac{\sqrt{3} + 3\sqrt{2}}{3}$$

e) i) $a = 31, d = 27 - 31$
 $d = -4$

$$T_n = a + (n-1)d$$

$$T_{30} = 31 + (30-1)(-4)$$

$$= -85$$

ii) $S_n = \frac{n}{2} [2a + (n-1)d]$

$$S_{50} = \frac{50}{2} [2(31) + (50-1)(-4)]$$

$$= -3350$$

$$a=2, b=5, c=-4$$

$$f) \alpha + \beta = -\frac{b}{a}$$

$$= -\frac{5}{2}$$

$$ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{-4}{2}$$

$$= -2$$

$$iii) (\alpha-1)(\beta-1)$$

$$= \alpha\beta - \alpha - \beta + 1$$

$$= \alpha\beta - (\alpha + \beta) + 1$$

$$= -2 - \left(-\frac{5}{2}\right) + 1$$

$$= \frac{3}{2}$$

$$9) i) y = x^2 - 9$$

$$y\text{-intercept: } x=0$$

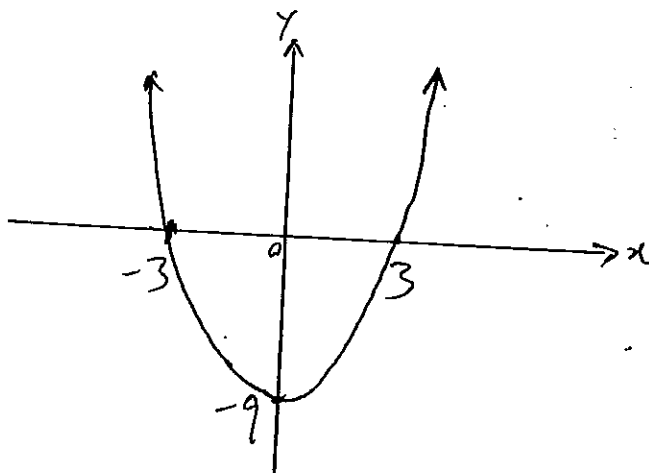
$$y = -9$$

$$x\text{-intercepts: } y=0$$

$$0 = x^2 - 9$$

$$x^2 = 9$$

$$x = \pm 3$$



$$ii) A = \left| \int_{-3}^3 x^2 - 9 \, dx \right|$$

$$= \left| \left[\frac{x^3}{3} - 9x \right]_{-3}^3 \right|$$

$$= 36 \text{ units}^2$$

Alternatively,

$$A = 2 \left| \int_0^3 x^2 - 9 \, dx \right|$$

$$= 2 \left| \left[\frac{x^3}{3} - 9x \right]_0^3 \right|$$

$$= 2 \left| [9 - 27] \right|$$

$$= 36 \text{ units}^2$$

$$12 \text{ a) i) } x^2 - 6x - 8y - 31 = 0$$

$$x^2 - 6x + 9 - 9 = 8y + 31$$

$$(x-3)^2 = 8y + 40$$

$$(x-3)^2 = 8(y+5) \rightarrow \text{GIVEN}$$

$$\text{ii) } 8 = 4a$$

$$a = 2$$

$$\text{focal length} = 2$$

$$\text{iii) Focus} = (3, -3)$$

$$\text{iv) Eqn of directrix:}$$

$$\text{v) } y = \frac{(x-3)^2}{8} - 5$$

$$\frac{dy}{dx} = \frac{x-3}{4}$$

$$\text{At } x=11,$$

$$\frac{dy}{dx} = \frac{11-3}{4}$$

$$= 2$$

$$m_{\text{normal}} = -\frac{1}{2}$$

$$\text{Eqn of normal:}$$

$$y-3 = -\frac{1}{2}(x-11)$$

$$2y-6 = -x+11$$

$$x+2y-17=0$$

$$\text{b) } f'(x) = 4x - 5$$

$$f(x) = \int 4x - 5$$

$$f(x) = 2x^2 - 5x + C$$

$$6 = 2(1)^2 - 5(1) + C$$

$$C = 9$$

$$\therefore f(x) = 2x^2 - 5x + 9$$

$$\left. \begin{array}{l} y = -7 \\ \text{OR } y = \frac{x^2 - 6x + 9 - 40}{8} \\ = \frac{x^2 - 6x - 31}{8} \end{array} \right\} \text{cfm}$$

$$y' = \frac{x}{4} - \frac{3}{4}$$

$$\text{at } x=11, m_{\text{tangent}} = 2$$

$$c) i) m_{CD} = m_{AB}$$

$$\frac{y-2}{x-6} = \frac{3}{2-(-4)}$$

$$\frac{y-2}{x-6} = \frac{1}{2}$$

$$2y-4 = x-6$$

$$x-2y-2=0 \rightarrow \text{GIVEN}$$

$$ii) \text{ Line CD: } x-2y-2=0 \quad \text{--- (1)}$$

$$\text{Line AD: } x+y+4=0 \quad \text{--- (2)}$$

$$(2)-(1): 3y+6=0$$

$$y=-2$$

$$\text{Sub } y=-2 \text{ into (1)}$$

$$x-2(-2)-2=0$$

$$x=-2$$

$$D = (-2, -2) \rightarrow \text{GIVEN}$$

$$iii) AB = \sqrt{[2-(-4)]^2 + 3^2}$$

$$= \sqrt{45} \text{ units}$$

$$= 3\sqrt{5} \text{ units}$$

$$iv) d = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$$

$$= \frac{|(1)(-4)+0+(-2)|}{\sqrt{1^2+(-2)^2}}$$

$$= \frac{6}{\sqrt{5}}$$

$$= \frac{6\sqrt{5}}{5} \text{ units}$$

$$v) A = \frac{1}{2} (3\sqrt{5} + \sqrt{80}) \times \frac{6\sqrt{5}}{5}$$

$$= (3\sqrt{5} + 4\sqrt{5}) \times \frac{3\sqrt{5}}{5}$$

$$= \frac{7\sqrt{5} \times 3\sqrt{5}}{5}$$

$$= 21 \text{ units}^2$$

$$\text{OR } m_{AB} = \frac{3-0}{2-(-4)} = \frac{3}{6} = \frac{1}{2}$$

equation of CD // to AB

$$(y-2) = \frac{1}{2}(x-6)$$

$$2y-4 = x-6$$

$$0 = x-2y-2 \text{ GIVEN}$$

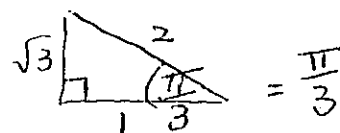
$$1.5 a) 4 \sin^2 2x = 3 \quad -\pi \leq x \leq \pi$$

$$\sin 2x = \pm \frac{\sqrt{3}}{2} \quad -2\pi \leq 2x \leq 2\pi$$

$$2x = +\frac{\pi}{3}, +\frac{2\pi}{3}, +\frac{4\pi}{3}, +\frac{5\pi}{3}$$

$$x = +\frac{\pi}{6}, +\frac{\pi}{3}, +\frac{2\pi}{3}, +\frac{5\pi}{6}$$

$$\pm 30^\circ, \pm 60^\circ, \pm 120^\circ, \pm 150^\circ$$



$$b) f(x) = x^3 + 6x^2 + 9x + 1$$

$$i) f'(x) = 3x^2 + 12x + 9$$

$$f''(x) = 6x + 12$$

$$\text{Stationary points: } f'(x) = 0$$

$$0 = 3x^2 + 12x + 9$$

$$0 = x^2 + 4x + 3$$

$$0 = (x+1)(x+3)$$

$$x = -1 \quad \text{or} \quad x = -3$$

$$y = -3$$

$$y = 1$$

$$f''(-1) = 6(-1) + 12$$

$$= -6 + 12$$

$$= 6$$

$$> 0$$

$$f''(-3) = 6(-3) + 12$$

$$= -18 + 12$$

$$= -6$$

$$< 0$$

$(-1, -3)$ is a ^{rel} minimum and $(-3, 1)$ is a ^{rel} maximum.

$$ii) \text{ Possible pt of inflexion: } f''(x) = 0$$

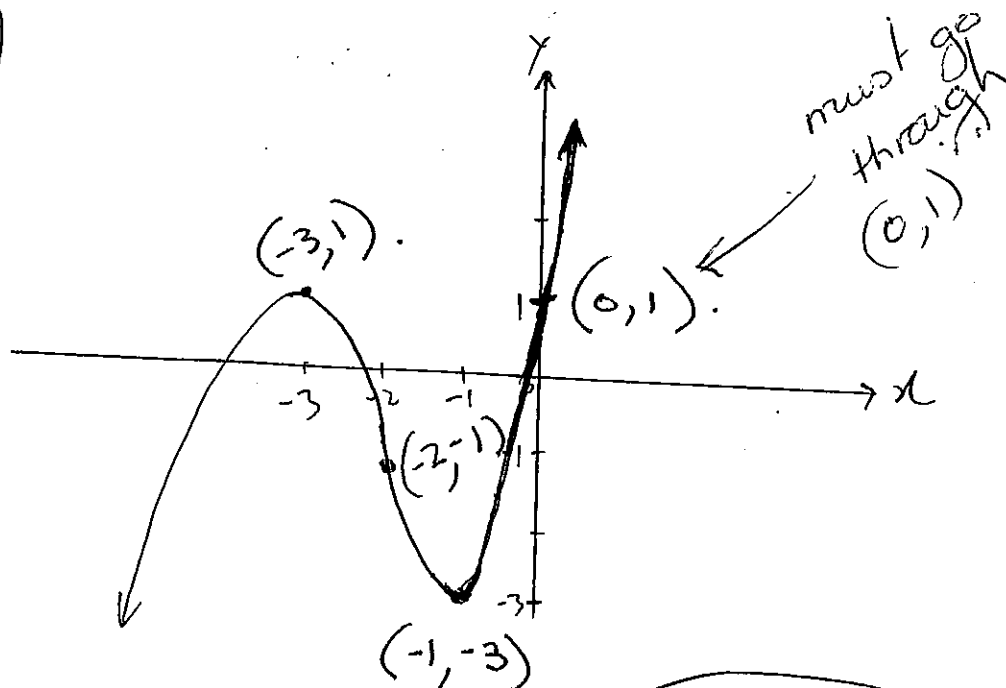
$$0 = 6x + 12$$

$$x = -2, y = -1$$

x	-3	-2	-1
$f''(x)$	-6	0	6

Concavity changes. $\therefore (-2, -1)$ is a point of inflexion.

iii)



c) i)

	y_0	y_1	y_2	y_3	y_4
x	1	2	3	4	5
$f(x)$	0	1.39	3.33	5.55	8.05

3.3

$$ii) \int_1^5 x \ln x \, dx = \int_1^3 x \ln x \, dx + \int_3^5 x \ln x \, dx$$

$$\approx \frac{3-1}{6} [0 + 4(1.39) + 3.33] + \frac{5-3}{6} [3.33 + 4(5.55) + 8.05]$$

OR $\frac{1}{3} \times \frac{5-1}{4} = 14.16$
 $[0 + 8.05 + 2(3.33) + 4(1.39 + 5.55)]$ (1 mark for 28.31 (cf)

d) $V = \pi \int_0^2 y^2 \, dx$

$$= \pi \int_0^2 x^4 \, dx$$

$$= \pi \left[\frac{x^5}{5} \right]_0^2$$

$$= \frac{32\pi}{5} \text{ units}^3 \text{ or } 6.4\pi$$

$$V = \pi \int_0^2 x^2 \, dx$$

$$= \frac{8\pi}{3}$$

OR 20.1

e) $A = \int_{-1}^2 e^{\frac{x}{2}} \, dx$

$$= 2 \left[e^{\frac{x}{2}} \right]_{-1}^2$$

$$= 2 \left[e - e^{-\frac{1}{2}} \right] =$$

$$\approx 0.79 \text{ (2 d.p.)}$$

$$= 4.22$$

Question 14

$$\begin{aligned} \text{a) i) } A_1 &= 50000 (1 + 0.06) - 4000 \\ &= 50000 (1.06) - 4000 \end{aligned}$$

$$A_2 = A_1 \times 1.06 - 4000$$

$$\begin{aligned} A_2 &= [50000 (1.06) - 4000] \times 1.06 - 4000 \\ &= 50000 (1.06)^2 - 4000 (1.06) - 4000 \\ &= 50000 (1.06)^2 - (4000 (1.06) + 4000) \quad \text{as required.} \end{aligned}$$

$$\begin{aligned} \text{ii) } A_3 &= A_2 (1.06) - 4000 \\ &= [50000 (1.06)^2 - (4000 (1.06) + 4000)] (1.06) - 4000 \\ &= 50000 (1.06)^3 - 4000 (1.06)^2 - 4000 (1.06) - 4000 \\ &= 50000 (1.06)^3 - 4000 (1.06^2 + 1.06 + 1) \end{aligned}$$

$$A_n = 50000 (1.06)^n - 4000 (1.06^{n-1} + 1.06^{n-2} + \dots + 1.06 + 1)$$

is a GP with $a=1$ $r=1.06$
n terms

$$\begin{aligned} A_n &= 50000 (1.06)^n - 4000 \left[1 \times \frac{1.06^n - 1}{1.06 - 1} \right] \\ &= 50000 (1.06)^n - \frac{4000 (1.06^n - 1)}{0.06} \quad \text{as required.} \end{aligned}$$

$$\text{iii) } A_{12} = 50000 (1.06)^{12} - \frac{4000 (1.06^{12} - 1)}{0.06}$$

$$\begin{aligned} &= \$33130.06 \\ &= \$33130 \quad (\text{nearest dollar}) \end{aligned}$$

$$\begin{aligned}
 \text{b) i) } \log_m(ab) &= \log_m a + \log_m b \\
 &= 1.2 + 0.5 \\
 &= 1.7
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \log_m \frac{b^3}{a} &= \log_m b^3 - \log_m a \\
 &= 3 \log_m b - \log_m a \\
 &= 3(0.5) - 1.2 \\
 &= 0.3
 \end{aligned}$$

$$\begin{aligned}
 \text{c) i) } \frac{1}{2x-3} - \frac{1}{2x+3} &= \frac{1(2x+3) - 1(2x-3)}{(2x-3)(2x+3)} \\
 &= \frac{2x+3-2x+3}{4x^2-9} \\
 &= \frac{6}{4x^2-9}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \int \frac{dx}{4x^2-9} &= \frac{1}{6} \int \frac{6}{4x^2-9} dx \\
 &= \frac{1}{6} \int \frac{1}{2x-3} - \frac{1}{2x+3} dx \\
 &= \frac{1}{6} \times \frac{1}{2} \int \frac{2}{2x-3} - \frac{2}{2x+3} dx \\
 &= \frac{1}{12} [\ln |2x-3| - \ln |2x+3|] + C \\
 &= \frac{1}{12} \ln \left| \frac{2x-3}{2x+3} \right| + C
 \end{aligned}$$

In $\triangle ADE$ and $\triangle EFB$

d) i) $\angle DEA = \angle FBE$ (corresponding angles on $ED \parallel BC$ are equal)

$\angle DAE = \angle FEB$ (corresponding angles on $AC \parallel EF$ are equal)

$\therefore \triangle ADE \parallel \triangle EFB$ (equiangular)

NOTE: Above is the most efficient solution but other solutions supported by valid reasons were accepted.

ii) $\frac{AD}{DE} = \frac{EF}{BF}$ (matching sides in $\triangle ADE \parallel \triangle EFB$ are in proportion)

$$\frac{2}{x} = \frac{x}{12-x}$$

$$24 - 2x = x^2$$

$$x^2 + 2x - 24 = 0$$

$$(x+6)(x-4) = 0$$

$$x = -6, 4$$

But x is a length $\therefore x > 0$ Hence $x = 4$.

Note: Students who used the ratio

$$\frac{2}{x} = \frac{2+x}{12}$$

gained 1 mark if they correctly solved for x , but did not gain credit as they had not established $\triangle ADE \parallel \triangle ACF$

$$15) a) i) \frac{dV}{dt} = -\frac{1600}{(t+4)^2}$$

$$= -1600(t+4)^{-2}$$

$$V = \int -1600(t+4)^{-2} dt$$

$$= 1600(t+4)^{-1} + C$$

$$V = \frac{1600}{t+4} + C$$

When $t=0$, $V=500$.

$$500 = \frac{1600}{4} + C$$

$$C = 100$$

$$\therefore V = \frac{1600}{t+4} + 100$$

When $t=20$,

$$V = \frac{1600}{20+4} + 100 \quad \text{or } \frac{500}{3} \text{ L}$$

$$= 166\frac{2}{3} \text{ litres}$$

ii) When $t \rightarrow \infty$,

$$V \rightarrow 100 \quad \text{as } \frac{1600}{t+4} \rightarrow 0$$

The volume will eventually be reduced to 100L.

b) i) $P = Ae^{kt}$

$$\frac{dP}{dt} = Ake^{kt}$$

$$= k(Ae^{kt})$$

$$= kP$$

ii) $P = 1.8A$ when $t=5$

$$1.8A = Ae^{5k}$$

$$\ln 1.8 = 5k$$

$$k = \frac{\ln 1.8}{5}$$

$$\approx 0.1176 \quad (4 \text{ sig. fig.})$$

$$\text{iii) } P = Ae^{0.1176t}$$

$$P > 100A$$

$$Ae^{0.1176t} > 100A$$

$$0.1176t > \ln 100$$

$$t > \frac{\ln 100}{0.1176}$$

$$t > 39.1696 \dots$$

$$t = 40$$

It will take 40 months.

39.16 or 39.2 also accepted.

$$\text{c) i) } a = 0.$$

$$\text{ii) } t = 2$$

iii) It represents the distance travelled ^{or displacement} in the first two seconds.

$$\text{d) i) } \dot{x} = 0$$

$$0 = 12(t - t^3)$$

$$0 = t(1 - t^2)$$

$$0 = t(1 + t)(1 - t)$$

$$t = 0, -1, 1$$

The particle comes to rest again after 1 second.

$$\text{ii) Distance travelled} = \left| \int_0^1 12(t - t^3) dt \right| + \left| \int_1^2 12(t - t^3) dt \right|$$

$$= \left| 12 \left[\frac{t^2}{2} - \frac{t^4}{4} \right]_0^1 \right| + \left| 12 \left[\frac{t^2}{2} - \frac{t^4}{4} \right]_1^2 \right|$$

$$= 12 \left(\frac{1}{2} - \frac{1}{4} \right) + \left| 12 \left[(2 - 4) - \left(\frac{1}{2} - \frac{1}{4} \right) \right] \right|$$

$$= 3 + 27$$

$$= 30 \text{ m}$$

or
Finding displacement function and constant of integration.

$$16) a) i) \frac{d}{dx} \left(\tan \frac{x}{3} \right) = \frac{1}{3} \sec^2 \left(\frac{x}{3} \right)$$

$$ii) \frac{d}{dx} (\sin^4 x) = 4 \sin^3 x \cos x$$

$$b) \int \sec^2 \left(\frac{x-3}{2} \right) dx = 2 \tan \left(\frac{x-3}{2} \right) + C$$

$$c) i) y = \sin x^2$$

$$y' = 2x \cos x^2$$

$$ii) \int x \cos x^2 dx = \frac{1}{2} \int 2x \cos x^2 dx$$

$$= \frac{1}{2} \sin x^2 + C$$

$$2 \int x \cos x^2 dx$$

$$\frac{1}{2} \sin x^2 + C$$

$$d) A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \cos x - \sin 2x dx$$

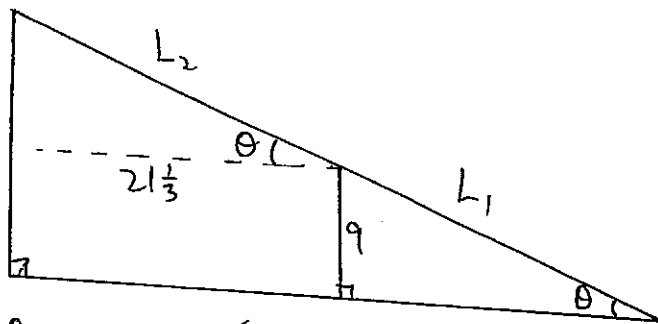
$$= \left[\sin x + \frac{\cos 2x}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{6}}$$

$$= \left[\left(\sin \frac{\pi}{6} + \frac{\cos \frac{\pi}{3}}{2} \right) - \left(\sin \left(-\frac{\pi}{2} \right) + \frac{\cos(-\pi)}{2} \right) \right]$$

$$= \frac{1}{2} + \frac{1}{4} - \left(-1 - \frac{1}{2} \right)$$

$$= 2 \frac{3}{4} \text{ units}^2$$

e) i)



$$\sin \theta = \frac{9}{L_1} \quad \cos \theta = \frac{21\frac{1}{3}}{L_2}$$

$$L_1 = \frac{9}{\sin \theta}$$

$$L_2 = \frac{21\frac{1}{3}}{\cos \theta}$$

$$L_2 = \frac{64}{3 \cos \theta}$$

$$L = L_1 + L_2$$

$$= \frac{9}{\sin \theta} + \frac{64}{3 \cos \theta}$$

ii)

$$L = 9(\sin \theta)^{-1} + \frac{64}{3}(\cos \theta)^{-1}$$

$$\frac{dL}{d\theta} = -9(\sin \theta)^{-2} \cos \theta - \frac{64}{3}(\cos \theta)^{-2}(-\sin \theta)$$

$$= -\frac{9 \cos \theta}{\sin^2 \theta} + \frac{64 \sin \theta}{3 \cos^2 \theta}$$

$$\frac{dL}{d\theta} = 0, \quad 0 = -\frac{9 \cos \theta}{\sin^2 \theta} + \frac{64 \sin \theta}{3 \cos^2 \theta}$$

$$27 \cos^3 \theta = 64 \sin^3 \theta$$

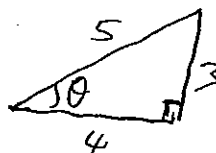
$$\tan^3 \theta = \frac{27}{64}$$

$$\tan \theta = \frac{3}{4}$$

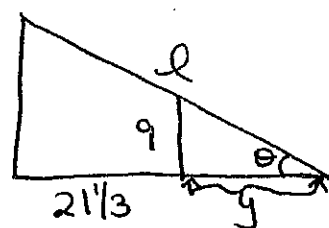
$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

θ	35°	$\tan^{-1}\left(\frac{3}{4}\right)$	40°
$\frac{dL}{d\theta}$	-0.921	0	1.620

Minimum $L = \frac{9}{\sin \theta} + \frac{64}{3 \cos \theta}$
 $= 9 \times \frac{5}{3} + \frac{64 \times 5}{3 \times \frac{4}{5}}$



$$\rightarrow = 41\frac{2}{3} \text{ m}$$



$$\tan \theta = \frac{9}{y}$$

$$y = \frac{9}{\tan \theta}$$

$$\cos \theta = \frac{21\frac{1}{3} + y}{l}$$

$$l = \frac{21\frac{1}{3} + y}{\cos \theta}$$

$$l = \frac{1}{\cos \theta} \left[21\frac{1}{3} + y \right]$$

$$= \frac{1}{\cos \theta} \left[\frac{64}{3} + \frac{9}{\tan \theta} \right]$$

$$= \frac{64}{3 \cos \theta} + \frac{9}{\sin \theta \cos \theta}$$

$$= \frac{64}{3 \cos \theta} + \frac{9}{\sin \theta}$$